SURV 622/SURVMETH 622: WEB SCRAPING AND APIs

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What are web scraping and APIs?

- **Web scraping** involves using a program to download and process web pages directly, transforming unstructured data into structured data that can be stored and analyzed
- **Application Programming Interfaces (APIs)** are tools provided by the owners of a website that enable scripted access to the data they hold
  - APIs are increasingly common
  - Using an API generally will be easier than scraping the same information
- Goal for today to help you understand what these tools can do
  - Will walk through an example of each in some detail, but are not trying to make you expert web scrapers or API programmers

Web-scraping to obtain data

Examples of data that might be scraped for statistical purposes

- Information on product prices and characteristics
  - Billion Prices Project (BPP) scrapes data to produce price indexes for 20 countries
  - Bureau of Labor Statistics scrapes data on prices and product characteristics to use in hedonic models for quality adjustment
- Lists of and information about businesses or organizations
  - Energy Information Administration has tested web-scraping to develop a survey frame of local governments that use alternative fuel vehicles
- Lists of and information about people in certain professions
  - State registries (for example, click here) or health insurance company lists of associated providers (for example, click here) could be scraped to develop survey sampling frames of medical service providers
  - A Maryland Economics graduate student recently scraped information for a project from the Maine registry of Certified Nursing Assistants (CNAs) (click here)

Web-scraping basics

- Identify website that contains data or information of interest
- Look at the web page to determine how the information is stored on the page
  - Right click and select inspect or enter <ctrl> U in browser to see the HTML code for a page
- Write code to extract the data of interest
- Once data obtained, configure them for analysis
- Challenges to obtaining web scraped data
  - Because each web page is unique, query will differ from page to page (must be customized)
  - Query must be modified if page format changes
  - Web pages often designed in ways that make scraping difficult
  - Website owner may impose restrictions on use

General Workflow

- Go to the website and check HTML.
- Figure out where the information is.
- Use R (or your web scraping tool of choice) to read the HTML.
- Use the tags that you found from inspection to isolate the data you want.
- Put data into a usable format (e.g. a dataframe).

XPath

- Selector Gadget: https://selectorgadget.com

Terms of service restrictions on web-scraping

- Many web sites include restrictions on scraping in their posted terms of service
  - Example: "**7. Prohibited Conduct**… (Y)ou may not…engage in unauthorized "scraping" or spidering, or harvesting of personal information, or use any unauthorized automated means to compile information." (Washington Post website)
  - Example: "Macy's grants you a limited, non-exclusive, non-transferable license to access and make non-commercial use of this website. This license does not include… (e) any use of data mining, robots or similar data gathering and extraction tools." (Macy's website)
- Reasons owners may seek to limit web scraping include
  - Desire to preserve the commercial value of a company's data, information or other online resources
  - In cases where information consists of entries by users of the website, desire to ensure that it is not used in ways the users had not anticipated
  - Concern that scraping will interfere with website functionality

Terms of service restrictions on web-scraping (continued)

- Courts have ruled that terms of service are enforceable against competitors using data in ways courts view as improper, but have not always found for the plaintiff
  - Recent case decided by the Ninth Circuit (Oracle USA, Inc. v. Rimini Street, Inc.) involved scraping of the Oracle website
  - Primary statute at issue the California Computer Data Access and Fraud Act (CDAFA), which prohibits unauthorized taking of data from a computer, computer system, or computer network
  - Court ruling stated "We hold that taking data using a *method* prohibited by the applicable terms of use, when the taking itself generally is permitted, does not violate the CDAFA."
- Website owners could make breach of contract claims, but questionable that provisions contained in a clickwrap agreement presented to all users prior to download are enforceable (Neuburger 2018)
- Website owner can take steps to restrict access from IP addresses that appear to be violating terms of service, for example
  - Deny access to IP addresses that make too many requests
  - Deny access to IP addresses that end up on "honeypot" pages
- Polite to comply with restrictions contained in terms of service

Restrictions on web-scraping in robots.txt files

- Many websites contain a robots.txt file accessible from the root directory that specifies limits on content that users are permitted to access
- This code tells all robots they can visit all files:

```
User-agent: *
Disallow:
```

- This code tells all robots to stay out of the website:

```
User-agent: *
Disallow: /
```

- This code tells two robots to stay out of a specific file:

```
User-agent: Googlebot
User-agent: Badbot
Disallow: /private/
```

Restrictions on web-scraping in robots.txt files (contd)

- This code tells robots to slow down the rate at which they access a site. Can be interpreted either to mean number of seconds between successive visits or size of the time window during which the site can be visited once.

  User-agent: *

  Crawl-delay: 3

- Some examples of robots.txt files
  - www.bls.gov/robots.txt
  - www.census.gov/robots.txt
- Although adhering to restrictions stated in robots.txt is voluntary, website owner may take steps to restrict access from IP addresses that appear to be behaving inappropriately
- Polite to comply with restrictions contained in robots.txt files

Using APIs to obtain and provide data

API basics

- An API is a tool designed to facilitate communication with users of a service
  - APIs we will talk about today are RESTful services, a type of API that works directly over the web to transfer data
- Operation of an API similar to a function call in a programming language
  - Call a specific URL
  - Pass in some inputs (e.g., specifying desired variable)
  - Get back an output (e.g., JSON code that contains desired information for processing)
- Website owner will provide information to programmers about what resources are available through the API and limitations on accessing them
  - Information on available data elements and associated codes
  - Limitations on uses of information extracted via the API
  - Rate limits (e.g., number of requests per period of time, total number of records extracted)

API basics (continued)

- Programmers may write code in PHP, Python, Ruby, R, Java, Perl or some other language to access the API
  - Developer page for a website's API often will provide sample code
- For many APIs, there are convenient "wrappers" that makes the process of requesting and receiving information easier
  - May be able to call a routine in another program that generates the relevant HTTP code rather than having to write it directly

Census API

- See available APIs at https://www.census.gov/data/developers/data-sets.html
  - American Community Survey (ACS)
  - Decennial Census
  - More
- Need an API key, but it is easy to get and use.
- Request API key here: https://api.census.gov/data/key_signup.html
- Can use censusapi R package
  - See https://github.com/hrecht/censusapi for info on usage.

Examples of websites with APIs

- Formerly public Twitter APIs
  - Search API searches for recent tweets (approximately past week)
  - Stream API collects tweets as they happen
  - Both have filtering capability
  - Both return only public tweets and have rate limits
- Several different LinkedIn APIs
  - APIs tap into information from LinkedIn member profiles; domains include people, organizations, and jobs
  - Potential applications include apps for adding particular credentials to LinkedIn profiles, facilitating contacts among LinkedIn members, and managing companies' social media presence on LinkedIn
  - Prohibits use for applications that compete with LinkedIn services, storage of information about individual members obtained using the APIs without their consent
- Indeed.com API
  - Indeed aggregates job listings from "thousands of company websites and job boards"
  - API accesses information contained in those listings

Examples of websites with APIs (cont’d)

- Bureau of Labor Statistics API
  - Allows user to request single or multiple series, specify desired years. Selected sample R code:

```
+#' ## Single Series request
+#' response <- blsAPI('LAUCN040010000000005')
+#' json <- fromJSON(response)

+#' ## Multiple Series
+#' payload <- list('seriesid'=c('LAUCN040010000000005', 'LAUCN040010000000006'))
+#' response <- blsAPI(payload)
+#' json <- fromJSON(response)

+#' ## One or More Series, Specifying Years
+#' payload <- list('seriesid'=c('LAUCN040010000000005', 'LAUCN040010000000006'),
'startyear'='2010', endyear'='2012')
+#' response <- blsAPI(payload)
+#' json <- fromJSON(response)
```

  - See here for examples of output from this type of call
  - Compare to existing BLS data access interface here

Other Examples of APIs

COVIDcast: https://cmu-delphi.github.io/delphi-epidata/api/covidcast.html

Global COVID-19 Trends and Impact Survey (CTIS):
https://gisumd.github.io/COVID-19-API-Documentation/